

WEB SCRAPING REPORT

Submitted By

Bhagyashree Pise

Table of Contents

1. Abstract
 2. Introduction
 3. Problem Statement
 4. Objectives
 5. Tools and Technologies
 6. Website Description
 7. Methodology
 8. Dataset Creation
 9. Data Analysis and Visualizations
 10. Key Findings
 11. Challenges Faced
 12. Conclusion
 13. Future Scope
-

1. Abstract

Web scraping is a powerful technique used to collect structured data from websites for analysis and decision-making. In this project, book-related information was extracted from a publicly available website using Python and BeautifulSoup. Multiple pages of the website were scraped to collect information such as book titles, prices, ratings, and availability status. The extracted data was stored in a structured dataset and analyzed using visualization techniques to identify patterns and trends.

2. Introduction

The internet contains a vast amount of valuable information that is often presented in an unstructured format. Web scraping enables automated extraction of such information and transforms it into structured datasets suitable for analysis.

This project demonstrates the practical implementation of web scraping by collecting book information from an online bookstore website. The extracted data was then analyzed to gain insights into book pricing and rating distributions.

3. Problem Statement

Organizations and analysts often require datasets that are not readily available in downloadable formats. Manual collection of data from websites can be time-consuming and inefficient.

The objective of this project is to automate the process of extracting book information from a website and create a structured dataset for further analysis.

4. Objectives

The objectives of this project are:

- Extract data from a public website using Python.
 - Understand website structure and HTML elements.
 - Navigate through multiple web pages.
 - Create a structured dataset from extracted information.
 - Perform exploratory analysis on the collected data.
 - Visualize patterns and trends using charts.
-

5. Tools and Technologies

The following tools and libraries were used:

Tool / Library	Purpose
Python	Programming Language
Requests	Sending HTTP Requests
BeautifulSoup	HTML Parsing and Data Extraction
Pandas	Data Manipulation and Analysis
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Jupyter Notebook	Development Environment

6. Website Description

Source Website

Website Used:

<http://books.toscrape.com>

The website is specifically designed for web scraping practice and contains detailed information about books including:

- Book Title
- Price
- Rating
- Availability

The website follows a paginated structure, making it suitable for demonstrating web navigation and multi-page scraping techniques.

7. Methodology

The project was completed in the following stages:

Step 1: Sending HTTP Requests

The Requests library was used to access web pages and retrieve HTML content.

Step 2: Parsing HTML Content

BeautifulSoup was used to analyze the HTML structure and locate relevant elements containing book information.

Step 3: Multi-Page Navigation

A loop was implemented to scrape data from the first five pages of the website.

Step 4: Data Extraction

The following attributes were extracted:

- Book Title
- Book Price
- Book Rating
- Availability Status

Step 5: Dataset Creation

The extracted information was stored in a Python list and converted into a Pandas DataFrame.

Step 6: Data Export

The final dataset was exported as a CSV file named:

books_dataset.csv

Step 7: Data Analysis

Statistical analysis and visualizations were performed to understand the characteristics of the collected data.

8. Dataset Creation

Dataset Attributes

Attribute	Description
Title	Name of the Book
Price	Book Price (£)
Rating	Rating Score (1–5)
Availability	Stock Availability

Sample Dataset

	Title	Price	Rating	Availability
0	A Light in the Attic	51.77	3	In stock
1	Tipping the Velvet	53.74	1	In stock
2	Soumission	50.10	1	In stock
3	Sharp Objects	47.82	4	In stock
4	Sapiens: A Brief History of Humankind	54.23	5	In stock

The final dataset contains information collected from multiple pages of the website and serves as the foundation for analysis.

9. Data Analysis and Visualizations

9.1 Basic Statistical Analysis

The following metrics were calculated:

- Total Books Scraped
- Average Price
- Highest Price
- Lowest Price
- Rating Distribution

Insert Output

```
Total Books Scraped: 100
Average Book Price: £34.56
Highest Book Price: £58.11
Lowest Book Price: £10.16

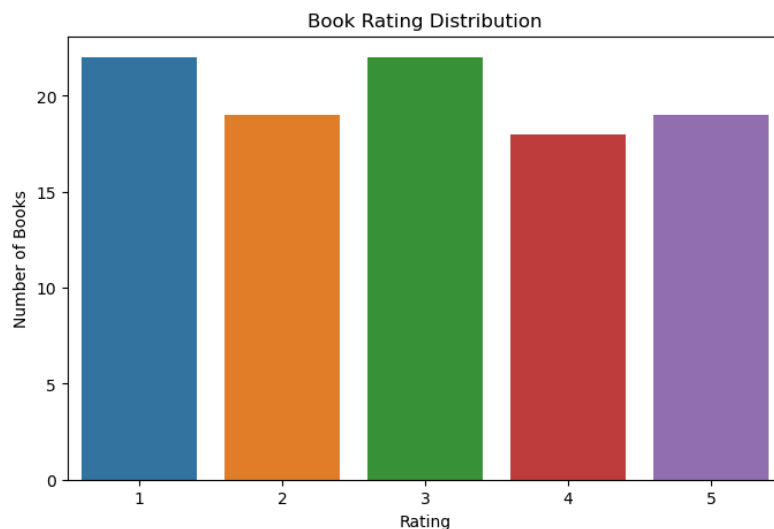
Most Common Rating:
1
```

```
Rating Distribution:
3    22
1    22
5    19
2    19
4    18
Name: Rating, dtype: int64
```

Observation

The statistical summary provides an overview of book pricing and rating patterns within the scraped dataset.

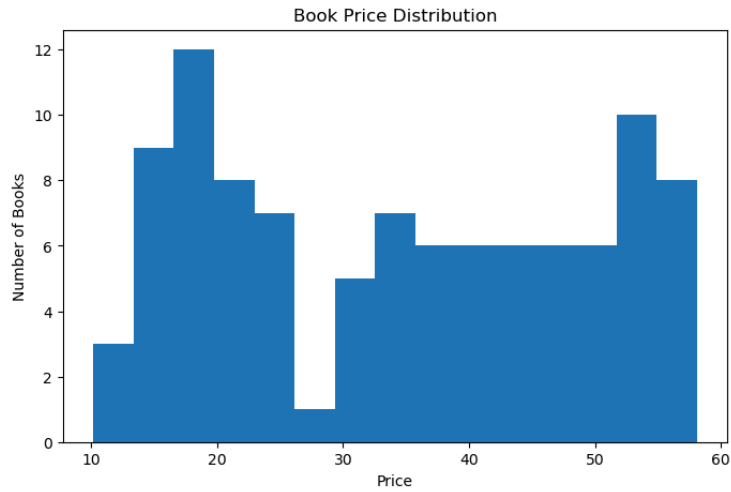
9.2 Book Rating Distribution



Observation

The chart illustrates how books are distributed across different rating categories. It helps identify the most common rating assigned to books.

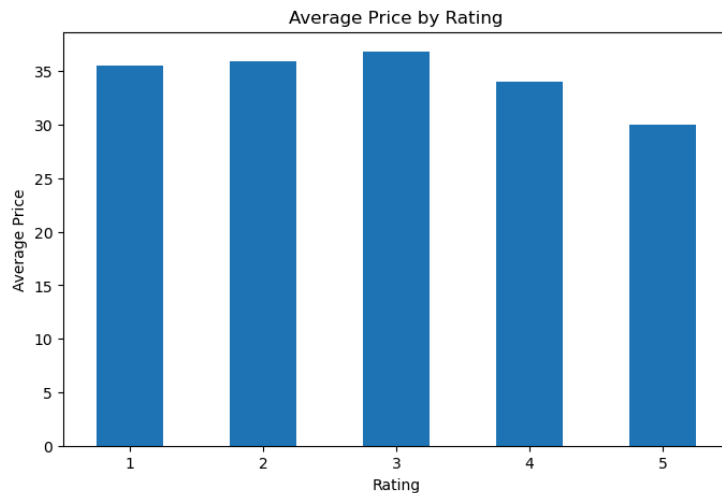
9.3 Book Price Distribution



Observation

The histogram shows the spread of book prices and highlights the price ranges where most books are concentrated.

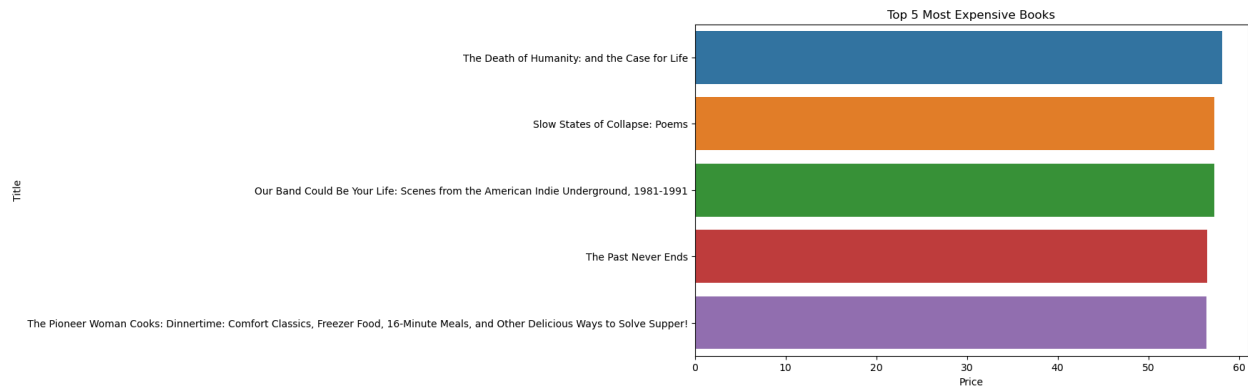
9.4 Average Price by Rating



Observation

This analysis compares average book prices across rating categories and helps identify whether highly rated books tend to have different pricing patterns.

9.5 Top 5 Most Expensive Books



Observation

The chart identifies the highest-priced books within the dataset and provides insight into premium-priced products.

10. Key Findings

The analysis produced several important findings:

- Data was successfully collected from multiple pages of the website.
 - A structured dataset was created automatically using web scraping techniques.
 - Book ratings varied across all categories from 1 to 5.
 - Price distribution revealed a range of low-cost and premium-priced books.
 - Visualization techniques made it easier to identify pricing and rating trends.
 - The project demonstrated the effectiveness of automated data collection compared to manual methods.
-

11. Challenges Faced

During the project, several challenges were encountered:

- Understanding the HTML structure of the website.
- Extracting specific elements using class names and tags.
- Cleaning extracted text values for numerical analysis.
- Navigating through multiple pages efficiently.
- Handling special characters present in price fields.

These challenges were resolved using BeautifulSoup, data cleaning techniques, and proper HTML inspection.

12. Conclusion

This project successfully demonstrated the complete workflow of web scraping using Python. Data was extracted from a public website, transformed into a structured dataset, and analyzed using statistical and visualization techniques.

The project highlights how web scraping can be used to create custom datasets that are not readily available for download. It also demonstrates the integration of data collection, cleaning, analysis, and visualization within a single workflow.

Overall, the project fulfilled all objectives and provided practical experience in web scraping and data analysis.

13. Future Scope

Future enhancements may include:

- Scraping all pages of the website instead of a limited number.
 - Collecting additional book information such as category and product description.
 - Implementing automated scheduling for data collection.
 - Building dashboards for interactive visualization.
 - Using advanced scraping frameworks such as Scrapy.
-

References

- [Python Documentation](#)
- [BeautifulSoup Documentation](#)
- [Pandas Documentation](#)
- [Matplotlib Documentation](#)
- [Seaborn Documentation](#)
- [Books to Scrape Website](#)